

Food Remedy API

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Assigned Task

This research investigates whether Food Remedy's actual application behaviour, data-handling processes and third-party SDK usage are consistent with its privacy and Google Play Data Safety statements.

The purpose is to identify what data the application collects, processes, stores or shares, compare this behaviour with release-facing privacy information, identify areas requiring further confirmation, and provide recommendations to keep the application's privacy disclosures aligned with its actual behaviour.

Summary

Food Remedy handles several types of user information through features such as account registration, user profiles, dietary preferences, product scanning, nutrition and allergen information, personalised recommendations, shopping lists and other application functions. The existing technical review confirms that the application uses **Firebase Authentication, Cloud Firestore, Firebase Storage and SQLite** as part of its data-handling environment.

This review found that privacy verification should not be limited to information directly entered by the user. Third-party services and SDKs also need to be considered because they may process technical information such as application identifiers, installation information or device metadata. Google Play requires Data Safety declarations to account for data collected or handled through third-party libraries and SDKs, not only data collected directly by the application's own code.

The main recommendation is therefore to maintain a **data-handling register** linking each Food Remedy feature and SDK to the type of data involved, purpose, storage location, sharing behaviour and retention/deletion approach. This information should then be checked against both the Privacy Policy and Google Play Data Safety form before each release.

1. Privacy and Data Safety Requirements

1.1 Google Play Data Safety

Google Play requires developers to provide information about how their application collects, shares and protects user data. Importantly, this responsibility extends to third-party libraries and SDKs included in the application.

For Google Play purposes, collection generally includes user data that is transmitted from the application off the user's device. Data handled only locally on the device, without being transmitted off-device, is treated differently. Google also distinguishes cases involving temporary or ephemeral processing.

1.2 Information That Should Be Verified

For each type of data used by Food Remedy, the following questions should be answered:

- What data is collected?
- Why is it required?
- Is it provided directly by the user or collected automatically?
- Does the data remain on the device?
- Is it transmitted to Firebase or another external service?
- Is it shared with another organisation or service?
- Is it processed only temporarily?
- How long is it retained?
- Can the user request deletion?
- Is the behaviour described in the Privacy Policy?
- Is it correctly represented in Google Play Data Safety?

This provides the basis for comparing the documented privacy statements with actual application behaviour.

2. Review of Food Remedy Data Handling

2.1 Account and Authentication Data

Food Remedy supports user registration, login and password recovery. The previous technical review identified **Firebase Authentication** as part of the application's authentication environment.

Potential data includes:

- Email address or another login identifier.
- Authentication account information.
- User identifier.
- Authentication/session information.
- Password-recovery information.

Because authentication involves a cloud service, this information should not simply be classified as local application data.

Privacy/Data Safety consideration: The Privacy Policy and Data Safety declaration should accurately identify account-related information that is transmitted to and processed through Firebase Authentication.

Status: Firebase Authentication use is confirmed; the exact production authentication methods and current release-facing disclosure require confirmation.

2.2 User Profile and Preference Data

The application supports user profiles containing demographic information and dietary preferences. It also provides personalised product insights.

Depending on the current implementation, this may include:

- User profile information.
- Dietary preferences.
- Allergy-related information.
- Food preferences or restrictions.
- Information used to personalise product recommendations.

This information deserves particular attention because dietary and allergy information can be sensitive from the user's perspective even where its formal regulatory classification depends on context and jurisdiction.

The review should determine whether this information remains on the device, is synchronised to Firestore, or is processed by another service.

Status: Profile and dietary functionality confirmed; exact fields, cloud storage and retention should be verified against the current implementation.

2.3 Product and Barcode Data

Food Remedy provides barcode scanning and product-search functionality.

A barcode itself generally identifies a product rather than the individual user. However, the privacy implications can change if product searches or scanned items are linked to a user's account, profile or shopping history.

The team should therefore confirm:

- Whether scan/search history is retained.
- Whether it is linked to a user identifier.
- Whether it is stored locally or remotely.
- Whether it is used to create personalised recommendations.

Status: Barcode scanning and product searching are confirmed features; retention and user-linking behaviour require verification.

2.4 Shopping, Cart and Application Activity

Food Remedy also supports shopping-list, cart and checkout functionality.

The review should determine what information from these functions is retained and whether it is stored locally or remotely. If shopping behaviour is associated with an identifiable account, this should also be considered when reviewing the application's privacy statements.

Status: Functionality confirmed; exact storage and retention behaviour requires confirmation.

2.5 Local SQLite Data

SQLite is used to support local/offline functionality, including caching and synchronisation.

Local storage should be distinguished from off-device collection. Google Play states that data accessed only on a user's device does not generally need to be declared as collected if it is never transmitted off the device.

However, Food Remedy includes synchronisation between Firebase and SQLite. Therefore, the team should not assume that information stored in SQLite is exclusively local. The data flow should be traced to determine whether individual fields are later transmitted to Firebase.

Status: SQLite and Firebase synchronisation are confirmed; field-level synchronisation requires further technical verification.

3. Third-Party SDK and Service Review

Third-party SDKs are important because their behaviour becomes part of the application's overall data-handling behaviour.

Google Play specifically states that developers are responsible for understanding what their SDKs collect, what permissions they use and why they use the data. Data collected by an SDK must be considered as part of the application's own Data Safety assessment.

3.1 Firebase

Firebase is currently the most important confirmed external data service for Food Remedy.

The project uses Firebase functionality including:

- Firebase Authentication.
- Cloud Firestore.
- Firebase Storage.

Firebase provides specific guidance for Google Play Data Safety because different Firebase SDKs can handle different categories of end-user information. Firebase also notes that actual disclosure depends on which SDKs are installed and how the application uses them.

For example, some Firebase components can collect technical information automatically. Firebase Installations generates a Firebase Installation ID and handles Firebase user-agent information. The latter can include device metadata such as OS version, device name/model/brand/form factor and information about Firebase SDK versions.

3.2 Expo / EAS

Expo is used as part of Food Remedy's React Native development and build environment. The project should distinguish between information handled by Expo during development/build activities and data handled by SDK functionality within the released application.

Expo provides its own data and privacy documentation explaining its treatment of developer and end-user information.

The production application should therefore be reviewed to establish which Expo modules are included and whether any relevant module introduces additional data collection or permissions.

Status: Expo/EAS use is confirmed; production SDK/module behaviour should be checked against the final application build.

3.3 Other SDKs and External Services

The final production dependencies should be reviewed using the project's package configuration and final Android build.

Attention should be given to SDKs associated with:

- Authentication.
- Storage.
- Analytics.
- Crash reporting.
- Notifications.
- Barcode scanning/camera access.
- External APIs.
- Web content.
- Payments or checkout, if applicable.

The presence of a library does not automatically mean that it collects personal data. Its actual configuration and use should be examined before determining whether a disclosure is required.

4. Data-Handling Comparison

A working comparison for Food Remedy can be maintained as follows:

Data / Service	Expected handling	Storage/processing	Current finding
Account/login data	Authentication	Firebase	Confirmed service; exact fields to verify
User profile	Personalisation/account	Firebase/local	Storage details to verify
Dietary preferences	Personalisation	Firebase/local	Handling to verify
Allergy information	Product suitability	Firebase/local	Handling and disclosure to verify
Barcode/product data	Product lookup	App/database	History retention to verify
Shopping/cart data	Shopping functionality	Local/cloud	Storage and retention to verify
Offline/cache data	Offline functionality	SQLite	Local handling confirmed; synchronisation to verify
Firebase technical data	Service operation	Firebase/Google	SDK-specific behaviour to verify
Expo modules	App/build functionality	Depends on module	Production modules to verify
Other SDKs	Depends on service	Depends on SDK	Final dependency review required

This table should be updated when the development team confirms the remaining technical details.

5. Potential Inconsistencies and Areas Requiring Confirmation

At this stage, the review does **not establish that the current Privacy Policy or Data Safety statement is incorrect**. Instead, it identifies areas where the documented statements need to be compared with actual application behaviour.

The main areas requiring confirmation are:

1. **Current Privacy Policy wording** – confirm that all current Food Remedy features and data categories are represented.
2. **Current Google Play Data Safety form** – obtain the existing declaration and compare each declared data category with the production application.
3. **Firebase data fields** – determine exactly which profile, preference and application data is transmitted to Firebase.
4. **Local versus cloud data** – determine which SQLite data remains exclusively on-device and which information is synchronised.
5. **Third-party SDK behaviour** – review the final production dependencies and identify any automatic or configuration-dependent data collection.
6. **Retention and deletion** – establish how long user information is retained and what happens when an account is deleted.
7. **Product activity/history** – confirm whether barcode searches, product interactions, shopping lists or cart activity are linked to identifiable user accounts and retained.
8. **Production-only services** – verify whether analytics, crash reporting, notification or other SDKs are enabled in the production build.

6. Recommendations

- Maintain a data-handling register showing what data Food Remedy collects, why it is needed, where it is stored, and whether it is shared.
- Review the Privacy Policy and Google Play Data Safety declaration before each release to ensure they match actual application behaviour.
- Regularly review Firebase and other third-party SDKs, particularly when SDKs are added or updated.
- Clearly distinguish between data stored locally, sent to Firebase, processed temporarily, or shared with external services.
- Document clear data retention and deletion procedures, including what happens when a user deletes their account.
- Include privacy verification in the release checklist, especially when new features, permissions or third-party services are introduced.
- Where data-handling behaviour is unclear, confirm it with the development team before making or updating privacy disclosures.